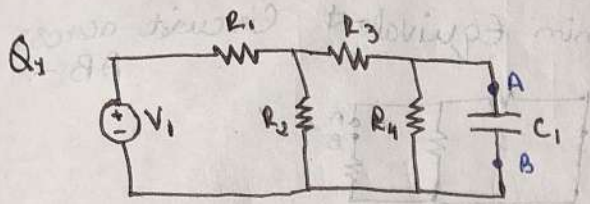
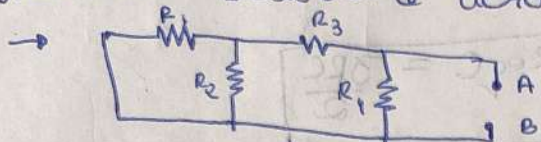


Name → Mukul Goel
Roll No. → 220108037

EE206 - Analog Circuits Assignment I



Thevenin Resistance across AB



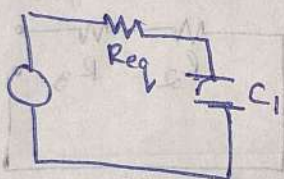
$$R_{eq} = ((R_1 \parallel R_2) + R_3) \parallel R_4$$

$$= \left(\frac{R_1 R_2}{R_1 + R_2} + R_3 \right) \parallel R_4$$

$$= \left(\frac{R_1 R_2 + R_1 R_3 + R_2 R_3}{R_1 + R_2} \right) \parallel R_4$$

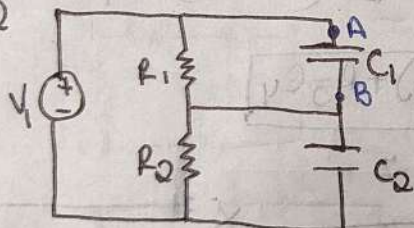
$$\Rightarrow R_{eq} = \frac{R_4 (R_1 R_2 + R_1 R_3 + R_2 R_3)}{R_1 R_2 + R_1 R_3 + R_2 R_3 + R_1 R_4 + R_2 R_4}$$

$$\therefore V_{Th} =$$

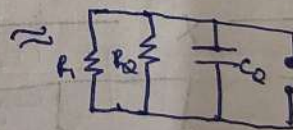
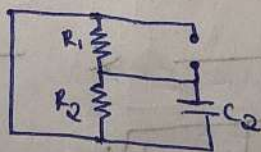


$$\Rightarrow V = R_{eq} C_1 = \frac{R_4 C_1 (R_1 R_2 + R_1 R_3 + R_2 R_3)}{R_1 R_2 + R_1 R_3 + R_2 R_3 + R_1 R_4 + R_2 R_4}$$

Q₂

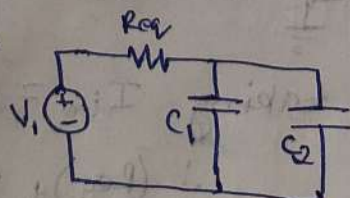


Thevenin Resistance across AB



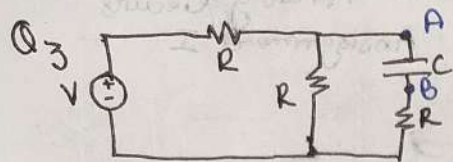
$$\Rightarrow R_{eq} = \frac{R_1 R_2}{R_1 + R_2}$$

⇒ Circuit Equivalent →

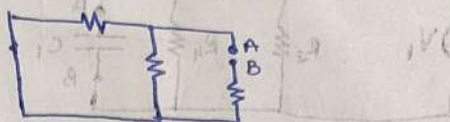


$$C_{eq} = C_1 + C_2$$

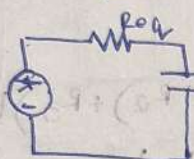
$$\tau = R_{eq} C_{eq} = \left(\frac{R_1 R_2}{R_1 + R_2} \right) (C_1 + C_2)$$



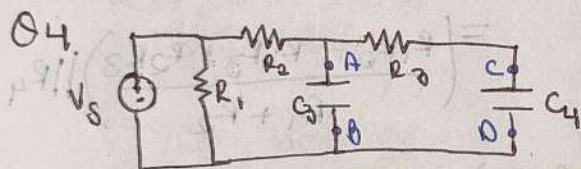
Thevenin Equivalent Circuit across AB -



$$R_{eq} = \frac{R \times R}{R + R} + R = \frac{3R}{2}$$



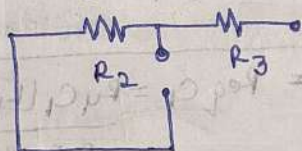
$$\tau = R_{eq} C = \frac{3RC}{2}$$



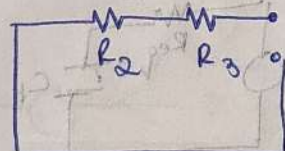
Thevenin Equivalent Ckt

With respect to AB

With respect to C

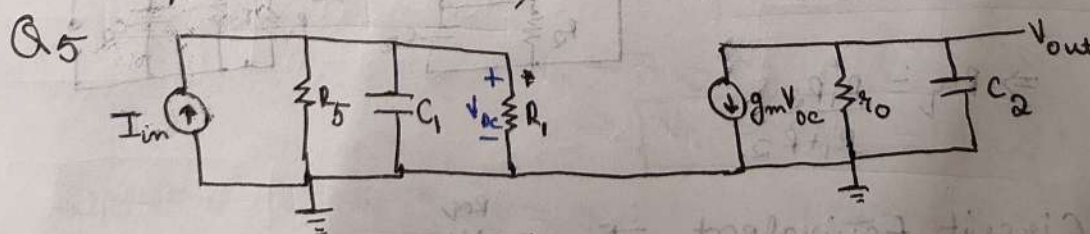


$$\tau_1 = R_2 C_3$$



$$\tau_2 = (R_2 + R_3) C_4$$

$$\Rightarrow \tau_{eq} = \tau_1 + \tau_2 = R_2 (C_3 + C_4) + R_3 C_4$$



For C_1 taking $I_{in} = 0$, we get $V_{oc} = 0$

$$\therefore (R_{eq})_1 = R_5 \parallel R_1$$

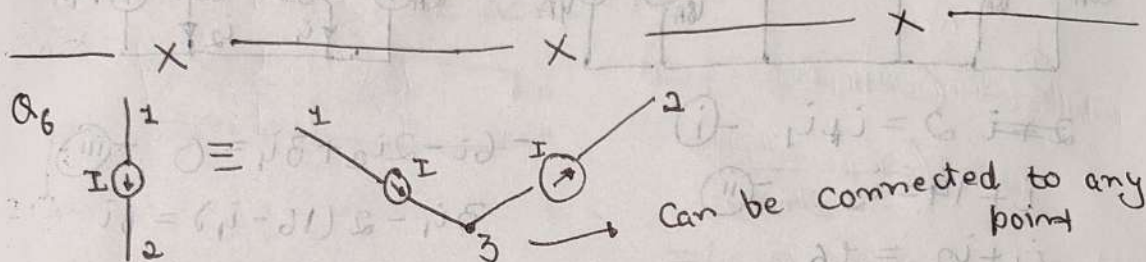
For C_2 taking $g_m V_{oc} = 0$

$$\therefore (R_{eq})_2 = r_o$$

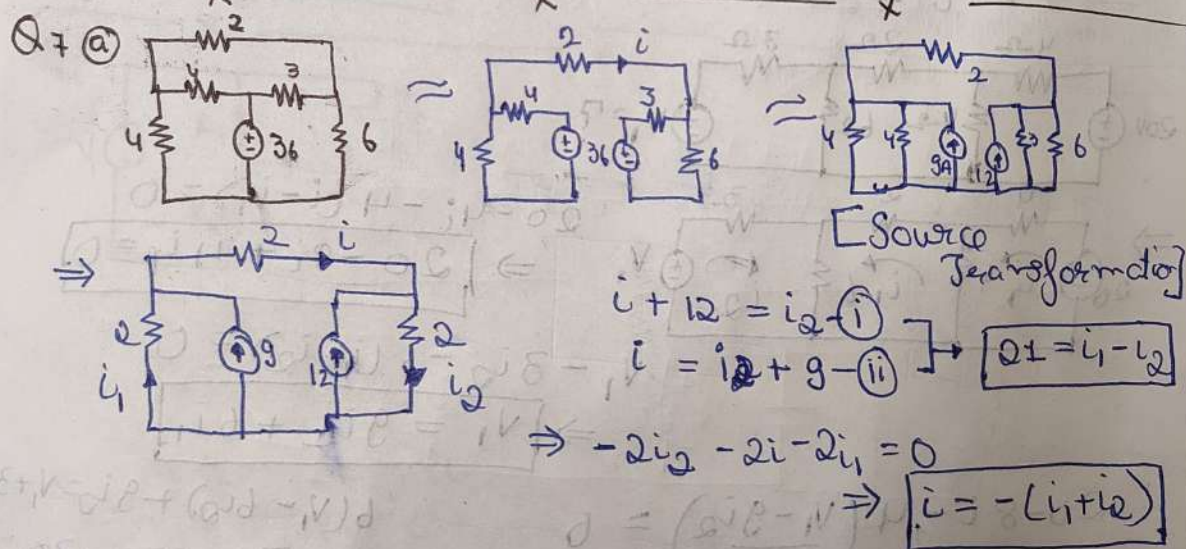
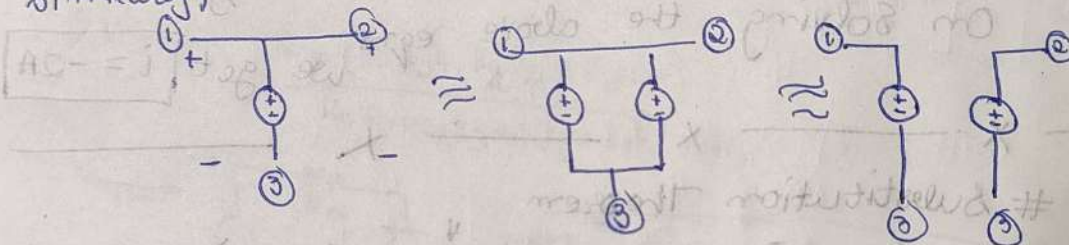
$$\therefore \tau_1 = (R_{eq})_1 C_1 = \left(\frac{R_5 R_1}{R_5 + R_1} \right) C_1$$

$$\therefore \tau_2 = (R_{eq})_2 C_2 = (R_0) C_2$$

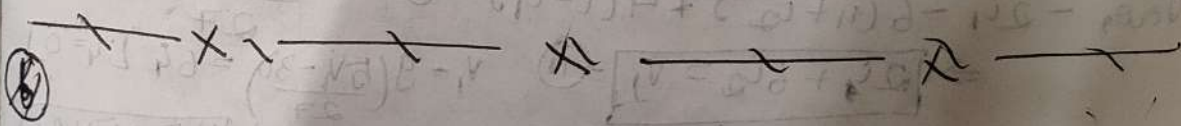
$$\tau_{eq} = \tau_1 + \tau_2 = \left(\frac{R_5 R_1}{R_5 + R_1} \right) C_1 + R_0 C_2$$

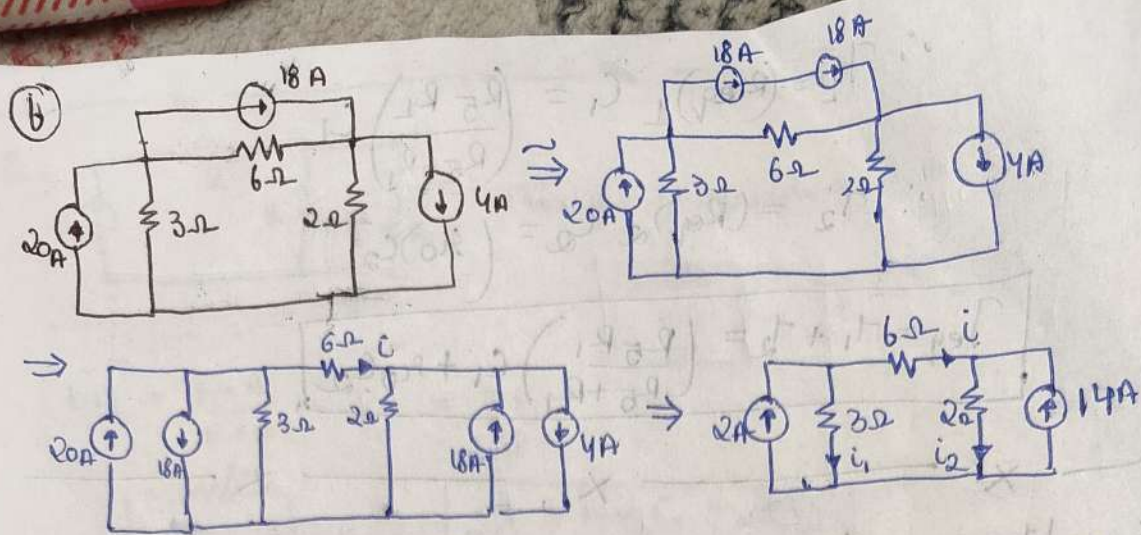


Similarly,



Solving the above eqns we get $i = \frac{2 \pm i}{2}$ $i = -1A$





$$20 = i + i_2 \quad \text{--- (i)}$$

$$i + 14 = i_2 \quad \text{--- (ii)}$$

$$i_1 + i_2 = 16$$

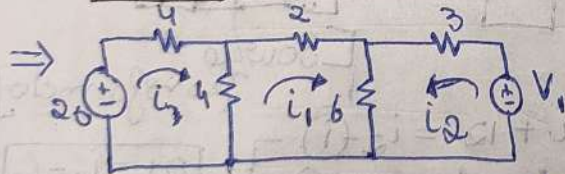
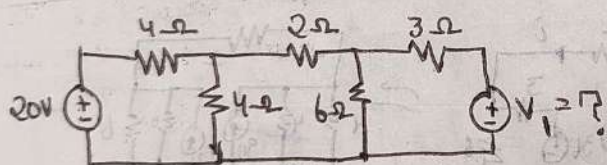
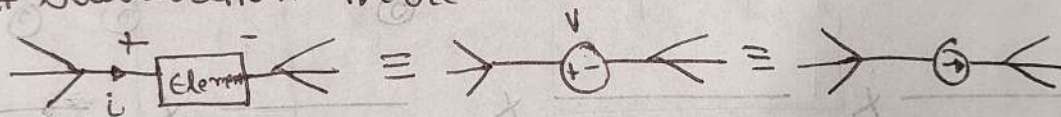
$$-6i - 2i_2 + 3i_1 = 0 \quad \text{--- (iii)}$$

$$3i_1 - 2(16 - i_1) = 6i$$

$$\Rightarrow i_1 = \frac{32 + 6i}{5}$$

On solving the above eqns we get $i = -2A$

Q8 # Substitution Theorem



$$20 - 4i_1 - 4(i_1 - i_2) = 0$$

$$\Rightarrow 20 - 8i_1 + 4i_2 = 0$$

$$V_1 - 3i_2 - 6(i_1 + i_2) = 0$$

$$\Rightarrow V_1 = 9i_2 + 6i_1$$

$$20 - 8i_1 - 4\left(\frac{V_1 - 9i_2}{6}\right) = 0$$

$$6(V_1 - 6i_2) + 9i_2 = V_1 + 30$$

$$\Rightarrow 4i_1 = 12i_1 + 9i_2 = V_1 + 30 \quad \text{--- (i)}$$

$$5V_1 - 27i_2 = 30$$

$$\text{Now, } -2i_1 - 6(i_1 + i_2) + 4(i_1 - i_2) = 0$$

$$5V_1 - 30 = i_2 \quad \text{--- (iii)}$$

$$\Rightarrow 2i_1 + 6i_2 = V_1 \quad \text{--- (ii)}$$

$$V_1 - 9\left(\frac{5V_1 - 30}{27}\right) = 64 \quad [i_1 = 0]$$

On solving the above eqn we get $V_1 = 15 \text{ Volts}$